Sports Science Research Update

May 2023

Cardiac arrest and neurologic recovery: Insights from the case of Mr. Damar Hamlin. Annal of Neurology, Ahead of Print: 1-13, 2023.

Effects of far-infrared radiation lamp therapy on recovery from muscle damage induced by eccentric exercise. European Journal of Sport Science, Ahead of Print: 1-10, 2023.

Effects of various numbers of runs on the success of hamstring injury prevention program in sprinters. International Journal of Environmental Research and Public Health, 19(9375): 1-13, 2023.

Exposures to near to maximal speed running bouts during different turnarounds in elite football: Association with match hamstring injuries. Biology of Sport, 40)4): 1057-1067, 2023.

Gluteal muscle forces during hip-focused injury prevention and rehabilitation exercises. Medicine and Science in Sports and Exercise, 55(4): 650-660, 2023.

Hamstring musculotendon mechanics of prospectively injured elite rugby athletes. Research in Sports Medicine, Ahead of Print: 1-13, 2023.

Nutritional strategies in the rehabilitation of musculoskeletal injuries in athletes: A systematic integrative review. Nutrients, 15(819): 1-25, 2023.

Prospective validation of 2B-Cool: Integrating wearables and individualized predictive analytics to reduce heat injuries. Medicine and Science in Sports and Exercise, 55(4): 751-764, 2023.

Public interest in cardiac arrest in the United States following Damar Hamlin's injury on Monday night football – Letter to the Editor. Resuscitation, Ahead of Print: 1-2, 2023.

Return to play, performance, and economic analysis of National Football League players after Lisfranc injury. The Orthopaedic Journal of Sports Medicine, 11(4): 1-7, 2023.

Return to sports following distal biceps tendon repair: A current concepts review. Journal of ISAKOS, Ahead of Print: 1-20, 2023.

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Safety of beta-alanine supplementation in humans: A narrative review. Sport Science for Health, Feb. 21: 1-7, 2023.

Telling the Whole Story: Bibliometric network analysis to evaluate impact of media attention on chronic traumatic encephalopathy research. Journal of Neurotrauma, 40: 148-154, 2023.

The Deceleration Index – Is it the missing link in rehabilitation? International Journal of Sports Physical Therapy, 18(2): 1-5, 2023.

The effect of cam and pincer morphology on hamstring injury rates and return to play in elite young male football players. Acta Orthopaedica et Traumatologica Turcica, 57(1): 36-39, 2023.

The effect of probable COVID-19 infection on the National Football League player's performance and endurance during the 2020 season. Cureus, 15(3): e35821: 1-6, 2023.

The good, the bad, and the ugly of consumer sleep technologies use among athletes: A call to action. Journal of Sport and Health Science, Ahead of Print: 1-3, 2023.

Validity of research based on publicly obtained data in sports medicine: A quantitative assessment of concussions in the National Football League. Sports Health, Ahead of Print: 1-10, 2023.

Sports Science Research Update

Cardiac arrest and neurologic recovery: Insights from the case of Mr. Damar Hamlin. Annal of Neurology, Ahead of Print: 1-13, 2023.

Abstract - The association between brain injury after cardiac arrest and poor survival outcomes has led to longstanding pessimism. However, the publicly witnessed cardiac arrest, resuscitation, and acute management of Mr. Damar Hamlin and his favorable neurologic recovery provides some optimism. Mr. Hamlin's case highlights the neurologic advances of the last 2 decades and presents the opportunity to improve outcomes for all cardiac arrest patients in key areas: (1) effectively implementing the American Heart Association "Chain of Survival" to prevent initial brain injury and promote neuroprotection; (2) revisiting the process of neurologic prognostication and re-defining the brain recovery during the early periods, and (3) incorporating neurorehabilitation into existing cardiac rehabilitation models to support holistic recovery.

Sports Science Research Update

Effects of far-infrared radiation lamp therapy on recovery from muscle damage induced by eccentric exercise. European Journal of Sport Science, Ahead of Print: 1-10, 2023.

Abstract - The present study investigated the effects of a far-infrared radiation (FIR) lamp therapy on changes in muscle damage and proprioception markers after maximal eccentric exercise of the elbow flexors (EF: Study 1) and the knee flexors (KF: Study 2) in comparison to a sham treatment condition. In each study, 24 healthy sedentary women were assigned to a FIR or a sham treatment group (n = 12/group). They performed 72 maximal EF eccentric contractions (Study 1) or 100 maximal KF eccentric contractions (Study 2) with their non-dominant limbs. They received a 30-min FIR (wavelength: 8–14 µm) or sham treatment at 1, 25, 49, 73 and 97 h postexercise to the exercised muscles. Maximum voluntary isometric contraction (MVC) torque, muscle soreness, plasma creatine kinase activity, and proprioception assessed by position sense, joint reaction angle, and force match were measured before, and 0.5, 24, 48, 72, 96 and 120 h post-exercise. The outcome measures showed significant changes ($P < 0.05$) at 0.5-hour postexercise (before treatment) similarly ($P > 0.05$) between the conditions in both studies. However, changes in all measures at 24–120 h post-exercise were smaller ($P < 0.05$) for the FIR than sham condition in both studies. For example, MVC torque returned to the baseline by 72 h postexercise for the FIR condition in both studies, but was still 19 ± 6% (Study 1) or 17 ± 12% (Study 2) lower than the baseline at 120 h post-exercise for the sham condition. These results suggested that the FIR lamp therapy was effective for accelerating recovery from muscle damage.

Highlights

- FIR lamp therapy was effective for recovering muscle strength and proprioception 1–3 days faster to the baseline.
- FIR lamp therapy was effective for attenuating muscle soreness by 55–60%, and reducing peak plasma CK activity by 45–89%.
- The effects of the FIR lamp therapy appear to be greater than other therapeutic interventions for eccentric exercise-induced muscle damage that have been investigated previously.

Sports Science Research Update

Effects of various numbers of runs on the success of hamstring injury prevention program in sprinters. International Journal of Environmental Research and Public Health, 19(9375): 1-13, 2023.

Abstract - Studies have not adequately addressed the influence of fatigue, which is considered a major risk factor for hamstring injuries. Therefore, this study aimed to clarify how a muscle fatigue condition affects the success of hamstring injury prevention programs in sprinters. The study subjects were 613 collegiate male sprinters. They employed submaximal/maximal running for a large number of runs and supramaximal running for a small number of runs in daily training. The hamstring injury prevention program had become the most effective strategy in the past 24 seasons of track and field for preventing hamstring injuries. The number of sprinters who experienced hamstring injuries in three periods over the 24 seasons was recorded. The incidents of hamstring injuries during supramaximal running per athlete-seasons were 137.9, 60.6, and 6.7 for Periods I, II, and III, respectively, showing a significant decline ($p < 0.01$). Furthermore, the incidents of hamstring injuries during submaximal and maximal running per season showed no significant change. The results of this study indicate that by inducing muscle fatigue, a small number of runs makes hamstring injury prevention programs effective.

Sports Science Research Update

Exposures to near to maximal speed running bouts during different turnarounds in elite football: Association with match hamstring injuries. Biology of Sport, 40)4): 1057-1067, 2023.

Abstract - To describe the occurrence of near-to-maximal sprinting speed (near-to-MSS) running bouts during training and hamstring injuries during the consecutive match of the same turnaround in elite football (soccer). Retrospective data from 36 team-seasons (16 elite teams performing in top European leagues) were analyzed (627 players, 96 non-contact time loss match hamstring injuries). We described 1) the occurrence of > 85%, > 90% or > 95% MSS exposures during training within each turnaround and match hamstring injuries and 2) whether the above-mentioned injury occurrences differed depending on the day(s) of the turnarounds (i.e., the period separating two consecutive matches, which is generally from 3 to 8 days) when these speed exposures occurred. The longer the length of the turnarounds and the lower the speed thresholds, the greater the number (and proportion) of near-to-MSS exposures (e.g., 18%, 45% and 72% of turnarounds with > 85% runs for 3, 5 and 7-turnarounds, respectively). For half of the turnarounds examined, there were no match hamstring injuries when players were exposed to running bouts > 95% MSS during training (e.g., injury rates: 0; CI: 0–15). Injuries still occurred during 85% of the turnarounds when there were no or lower relative speed exposures (i.e., > 85 or > 90%, injury rates: 2–5, CI: 0–6). Finally, irrespective of the turnaround length, there were no match hamstring injuries when > 95% MSS exposures occurred at D-2, while in contrast, injuries still happened when players were not exposed at all, or when these exposures occurred at D-3 and/or earlier within the turnaround. While the present observational study design precludes the examination of causal relationships, the programming of > 95% MSS exposures at D-2 may help mitigate match hamstring injury occurrences in elite football.

Sports Science Research Update

Gluteal muscle forces during hip-focused injury prevention and rehabilitation exercises. Medicine and Science in Sports and Exercise, 55(4): 650-660, 2023.

Purpose: This study aimed to compare and rank gluteal muscle forces in eight hip-focused exercises performed with and without external resistance and describe the underlying fiber lengths, velocities, and muscle activations.

Methods: Motion capture, ground reaction forces, and electromyography (EMG) were used as input to an EMG-informed neuromusculoskeletal model to estimate gluteus maximus, medius, and minimus muscle forces. Participants were 14 female footballers (18–32 yr old) with at least 3 months of lower limb strength training experience. Each participant performed eight hip-focused exercises (single-leg squat, split squat, single-leg Romanian deadlift [RDL], single-leg hip thrust, banded side step, hip hike, side plank, and side-lying leg raise) with and without 12 repetition maximum (RM) resistance. For each muscle, exercises were ranked by peak muscle force, and k-means clustering separated exercises into four tiers.

Results: The tier 1 exercises for gluteus maximus were loaded split squat (95% confidence interval [CI] = 495–688 N), loaded single-leg RDL (95% CI = 500–655 N), and loaded single-leg hip thrust (95% CI = 505–640 N). The tier 1 exercises for gluteus medius were body weight side plank (95% CI = 338–483 N), loaded single-leg squat (95% CI = 278–422 N), and loaded single-leg RDL (95% CI = 283–405 N). The tier 1 exercises for gluteus minimus were loaded single-leg RDL (95% CI = 267–389 N) and body weight side plank (95% CI = 272–382 N). Peak gluteal muscle forces increased by 28–150 N when exercises were performed with 12RM external resistance compared with body weight only. Peak muscle force coincided with maximum fiber length for most exercises.

Conclusions: Gluteal muscle forces were exercise specific, and peak muscle forces increased by varying amounts when adding a 12RM external resistance. These findings may inform exercise selection by facilitating the targeting of individual gluteal muscles and optimization of mechanical loads to match performance, injury prevention, or rehabilitation training goals.

Sports Science Research Update

Hamstring musculotendon mechanics of prospectively injured elite rugby athletes. Research in Sports Medicine, Ahead of Print: 1-13, 2023.

Abstract - The musculotendon mechanics of the hamstrings during high-speed running are thought to relate to injury but have rarely been examined in the context of prospectively occurring injury. This prospective study describes the hamstring musculotendon mechanics of two elite rugby players who sustained hamstring injuries during on-field running. Athletes undertook biomechanical analyses of high-speed running during a Super Rugby pre-season, prior to sustaining hamstring injuries during the subsequent competition season. The biceps femoris long head muscle experienced the greatest strain of all hamstring muscles during the late swing phase. When expressed relative to force capacity, biceps femoris long head also experienced the greatest musculotendon forces of all hamstring muscles. Musculotendon strain and force may both be key mechanisms for hamstring injury during the late swing phase of running.

Sports Science Research Update

Nutritional strategies in the rehabilitation of musculoskeletal injuries in athletes: A systematic integrative review. Nutrients, 15(819): 1-25, 2023.

Abstract - It is estimated that three to five million sports injuries occur worldwide each year. The highest incidence is reported during competition periods with mainly affectation of the musculoskeletal tissue. For appropriate nutritional management and correct use of nutritional supplements, it is important to individualize based on clinical effects and know the adaptive response during the rehabilitation phase after a sports injury in athletes. Therefore, the aim of this PRISMA in Exercise, Rehabilitation, Sport Medicine and Sports Science PERSiST-based systematic integrative review was to perform an update on nutritional strategies during the rehabilitation phase of musculoskeletal injuries in elite athletes. After searching the following databases: PubMed/Medline, Scopus, PEDro, and Google Scholar, a total of 18 studies met the inclusion criteria (Price Index: 66.6%). The risk of bias assessment for randomized controlled trials was performed using the RoB 2.0 tool while review articles were evaluated using the AMSTAR 2.0 items. Based on the main findings of the selected studies, nutritional strategies that benefit the rehabilitation process in injured athletes include balanced energy intake, and a high-protein and carbohydrate-rich diet. Supportive supervision should be provided to avoid low energy availability. The potential of supplementation with collagen, creatine monohydrate, omega-3 (fish oils), and vitamin D requires further research although the effects are quite promising. It is worth noting the lack of clinical research in injured athletes and the higher number of reviews in the last 10 years. After analyzing the current quantitative and non-quantitative evidence, we encourage researchers to conduct further clinical research studies evaluating doses of the discussed nutrients during the rehabilitation process to confirm findings, but also follow international guidelines at the time to review scientific literature.

Sports Science Research Update

Prospective validation of 2B-Cool: Integrating wearables and individualized predictive analytics to reduce heat injuries. Medicine and Science in Sports and Exercise, 55(4): 751-764, 2023.

Introduction: An uncontrollably rising core body temperature (TC) is an indicator of an impending exertional heat illness. However, measuring TC invasively in field settings is challenging. By contrast, wearable sensors combined with machine-learning algorithms can continuously monitor TC nonintrusively. Here, we prospectively validated 2B-Cool, a hardware/software system that automatically learns how individuals respond to heat stress and provides individualized estimates of TC, 20-min ahead predictions, and early warning of a rising TC.

Methods: We performed a crossover heat stress study in an environmental chamber, involving 11 men and 11 women (mean ± SD age = 20 ± 2 yr) who performed three bouts of varying physical activities on a treadmill over a 7.5-h trial, each under four different clothing and environmental conditions. Subjects wore the 2B-Cool system, consisting of a smartwatch, which collected vital signs, and a paired smartphone, which housed machine-learning algorithms and used the vital sign data to make individualized real-time forecasts. Subjects also wore a chest strap heart rate sensor and a rectal probe for comparison purposes.

Results: We observed very good agreement between the 2B-Cool forecasts and the measured TC, with a mean bias of 0.16°C for TC estimates and nearly 75% of measurements falling within the 95% prediction intervals of ±0.62°C for the 20-min predictions. The early-warning system results for a 38.50°C threshold yielded a 98% sensitivity, an 81% specificity, a prediction horizon of 35 min, and a false alarm rate of 0.12 events per hour. We observed no sex differences in the measured or predicted peak TC.

Conclusion: 2B-Cool provides early warning of a rising TC with a sufficient lead time to enable clinical interventions and to help reduce the risk of exertional heat illness.

Sports Science Research Update

Public interest in cardiac arrest in the United States following Damar Hamlin's injury on Monday night football – Letter to the Editor. Resuscitation, Ahead of Print: 1-2, 2023.

The only interventions known to improve survival following cardiac arrest are immediate high-quality cardiopulmonary resuscitation (CPR) and proper use of an automatic external defibrillator (AED).1 Damar Hamlin, a 24-year-old football player, experienced a cardiac arrest during a televised National Football League (NFL) game on January 2nd, 2023, after which athletic trainer, Denny Kellington, entered the field and immediately performed CPR.2 The speculation from news media and experts proposed commotio cordis—a condition where blunt force trauma disrupts the electrical activity of the heart— was likely the reasoning for Hamlin's cardiac arrest. Given that public knowledge and awareness of cardiac arrest and its emergent treatment is crucial for ensuring patient survival, we sought to determine public interest in related terms following Hamlin's injury and resuscitation.

We used Google Trends to extract daily search interest related to topics of "cardiac arrest," "cardiopulmonary resuscitation," and injury of "commotio cordis" within the United States for a 90-day time frame from October 12, 2022, through January 8, 2023, representing approximately 80 days prior to and 5 days after the event. Google Trends provides reliable data for tracking public interests, and the Google search engine accounts for a majority of internet queries performed in the United States.3 Search interest from Google Trends is reported as relative search interest (RSI) on a scale from 0-100, where 100 is the peak interest within the searched time frame rather than as absolute values. We identified the date(s) of peak RSI. Next, we constructed an auto-regressive integrated moving algorithm (ARIMA) to project search interest for each of these topics if the injury to Damar Hamlin had not occurred.

RSI for 'cardiac arrest,' 'cardiopulmonary resuscitation,' and 'commotio cordis' peaked on January 4, 2023 (Fig. 1). The difference in forecasted and actual RSI for 'cardiac arrest' was 99.01— a 10,023.46% increase above the expected value. Similar differences were seen in 'cardiopulmonary resuscitation' and 'commotio cordis.'

Sports Science Research Update

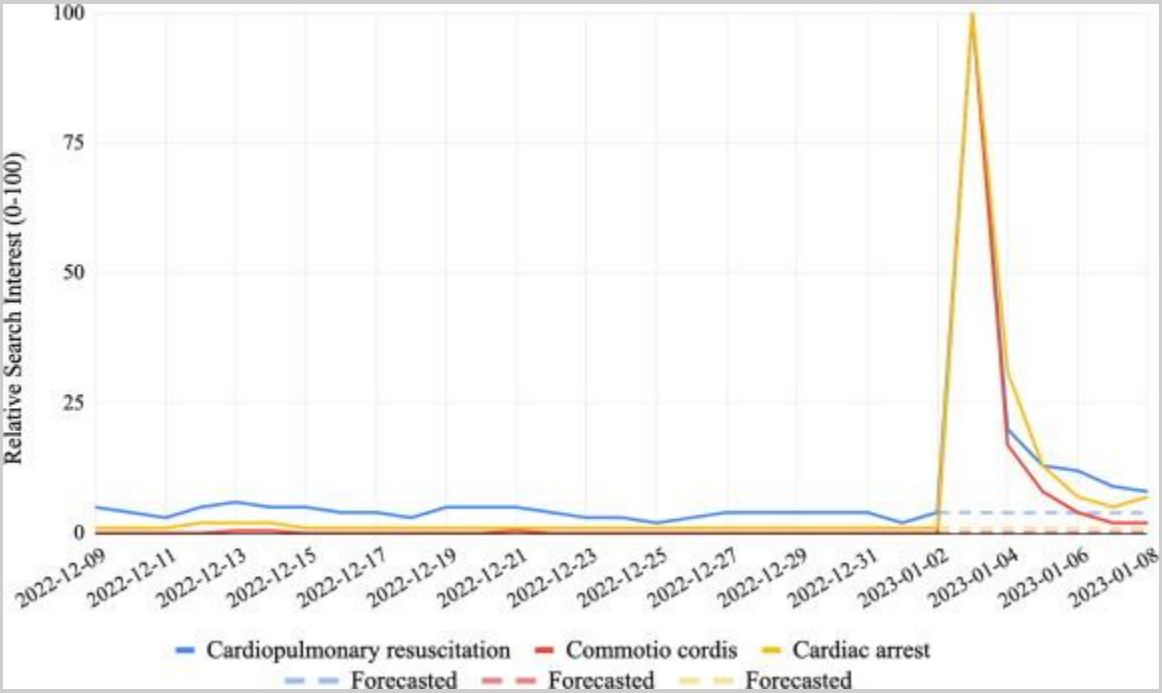

Fig. 1 Actual relative search volumes for each term following a NFL game in which Damar Hamlin survived a cardiac arrest are designated by the colored solid lines. The colored, dashed lines are the forecasted values as if Hamlin's injury did not occur from the autoregressive integrated moving average models for each term.

Following Hamlin's cardiac arrest and intervention by Kellington, our study showed that public interest significantly increased in CPR, cardiac arrest, and commotio cordis. Other studies have shown similar spikes in search interest in the US following US-based professional athletes sustaining orthopedic injuries.4 Following his recovery, Hamlin has partnered with the American Heart Association to engage and sustain interest in CPR training, which may allow further utilization of media to promote cardiac arrest intervention techniques and bridge the gaps in knowledge and confidence in action. Previous research noted a deficit in lifesaving skills within a large urban area showing only 19.7% of respondents stating, as a bystander, they would perform CPR when presented with a sudden cardiac arrest scenario,5 while only 7.9% of respondents mentioned the use of an AED.5 One training session alone can allow communities to gain a life skill that would better prepare them for emergent situations. With increased training, success of cardiac arrest recovery can improve on a populational level; thus, sustained media attention may be warranted.

Sports Science Research Update

Return to play, performance, and economic analysis of National Football League players after Lisfranc injury. The Orthopaedic Journal of Sports Medicine, 11(4): 1-7, 2023.

Background: A Lisfranc injury can occur to either the ligament or bone, which causes instability when attempting to perform strength and balance maneuvers.

Purpose/Hypothesis: The study's aims were to (1) analyze the return-to-play (RTP) rate and performance level of players in the National Football League (NFL) after Lisfranc injury and (2) determine the economic and financial impact of Lisfranc injuries to the NFL. We hypothesized that there would be a low RTP rate following Lisfranc injury in the NFL.

Study Design: Case series; Level of evidence, 4.

Methods: Multiple online public records were used to identify NFL players with Lisfranc injuries between the 2009 and 2020 seasons. Players were assessed according to their RTP statistics: snap count, approximate value (AV), games played, and games started. Nonparametric methods were used to compare player statistics before and after injury. Also, player salaries while injured were calculated (in 2022 US dollars after adjusting for inflation) to approximate economic loss for those players who returned to play.

Results: A total of 33 NFL athletes sustained a Lisfranc injury during the study period. Most Lisfranc injuries were experienced by offensive linemen (n = 8; 24.2%), followed by running backs (n = 7; 21.2%). Overall 27 players (81.8%) returned to play after injury at a median of 11.0 months (IQR, 10.2-11.8 months). There were no significant differences between pre- and postinjury snap counts or number of games played and started. In terms of player performance, there was a statistically significant decrease in AV at 1 year postinjury (median [IQR], 6.0 [4.0-10.0] preinjury vs 5.0 [2.5-7.5] postinjury; P = .022). The overall cost of recovery amounted to $104.7 million, with quarterbacks (n = 4) accounting for the greatest cost at $32.6 million. The next 2 most expensive positions were offensive and defensive linemen (n = 5 each) at $19.4 million each.

Conclusion: Our results did not support the hypothesis, as the RTP rate for NFL athletes sustaining Lisfranc injuries was 81.8%. This injury was associated with a significant decrease in AV 1-year postinjury. In terms of economic impact, quarterbacks accounted for almost one-third of expenses while constituting only 14.8% of injuries.

Sports Science Research Update

Return to sports following distal biceps tendon repair: A current concepts review. Journal of ISAKOS, Ahead of Print: 1-20, 2023.

Abstract - Distal biceps tendon ruptures are relatively rare injuries but tend to occur in active and athletic populations, especially in weightlifting and contact sports. The distal biceps tendon is an important supinator of the forearm and flexor of the elbow, thus an injury to this ligament can be invalidating for athletes. The aim of this review was to determine the ability and the time to return to sports following distal biceps tendon repair in athletes and the level of performance. The literature is scarce about the return to sports among athletes. Most studies include athletes are National Football League (NFL) players, others are weightlifters and a few recreational athletes. The return to play rate after distal biceps tendon repair is high. The performances of the returned players were similar to matched players and most players returned to the same level. Most players – depending on the sport – were not able to return to competition within the same season. In order to manage expectations, it should be discussed preoperatively with the athlete (and their coach) that the return to sports rate is high, but the return will probably be the following season.

Current Concepts - Distal biceps tendon repair has good clinical functional outcomes and the return-to-work rate is high. Following distal biceps repair, 95% of the athlete's tendon return to sports and 82% to the same preinjury level of competition. Functional outcomes following surgery are good and generally not clinically different compared to matched athletes. The time to return to sports is rather long, about 40 weeks in the current literature, depending on the type of sport.

Future perspectives - The current literature on return to sports following distal biceps injuries and repairs is sparse. Future research is necessary to compare surgical techniques, rehabilitation programs (such as early mobilization), and functional outcomes in order to draw conclusions about factors on return to sports.

Sports Science Research Update

Safety of beta-alanine supplementation in humans: A narrative review. Sport Science for Health, Feb. 21: 1-7, 2023.

Abstract - Carnosine levels play a significant role in intracellular pH buffering during exercise. The limiting factor for muscle carnosine synthesis is the availability of β-alanine. Results from meta-analysis showed a significant positive effect of β-alanine supplementation on high-intensity exercise performance. Nonetheless, much less has been described about the safety of β-alanine. The available literature indicates no adverse events related to β-alanine usage; side effects such as paresthesia may be observed if a large single high dose of β-alanine is taken, but the symptom can be attenuated either by using splitting doses (<1.6 g) or a sustained-release formula. No adverse effects have been reported up to 24 weeks of β-alanine supplementation (3.2 g.day−1), but the adverse effects of longer supplementation periods are still unknown. Most of the existing clinical studies were not designed with safety evaluations as primary end points. β-Alanine is currently considered to be safe in healthy populations, at recommended doses, by the International Society of Sports Nutrition stands position, while the Australian Institute of Sports has established β-alanine as a safe performance-enhancing supplement with strong scientific evidence (grade A). Cohort studies are needed evaluating the safety of β-alanine among representative populations, and the effects of co-variables such as sex, age, and ethnicity.

Sports Science Research Update

Telling the Whole Story: Bibliometric network analysis to evaluate impact of media attention on chronic traumatic encephalopathy research. Journal of Neurotrauma, 40: 148-154, 2023.

Abstract - There is a national debate regarding the existence of a relationship between contact sport participation and future risk of neurodegenerative disease. We employed bibliometrics and altmetrics to quantify the academic, popular, and social media impact of published scientific articles that report an association between contact sports or military service with chronic traumatic encephalopathy (CTE+), and compare with those scientific articles that report null or no association of contact sports or military service with CTE (CTE-). In this cross-sectional study, we extracted number of citations, total link strength, altmetric score, number of news stories, media outlets, and Twitter interaction from published CTE articles. The top 10 most cited articles were statistically compared on these outcomes using Mann-Whitney U tests. CTE+ publications had an average of 101 citations per article, Altmetric score of 272, 36 news stories in 26 media outlets, and upper-bound of Twitter users of 402,159. CTE- publications had an average of 29 citations per article, Altmetric score of 39, two news stories and media outlets, and upper-bound of Twitter users of 91,070. Top 10 CTE+ publications had, on average, 94% more citations ($p < 0.001$), 95% higher altmetric scores ($p = 0.01$), 99% higher number of news stories ($p = 0.01$), 98% higher number of media outlets ($p = 0.01$), and reached 95% more Twitter users than top 10 CTE- publications ($p = 0.11$). The bibliometric analysis indicates a significant inequality in media dissemination and popular consumption of scientific findings that do not support a relationship between contact sports or military service and future neurodegeneration.

Sports Science Research Update

The Deceleration Index – Is it the missing link in rehabilitation? International Journal of Sports Physical Therapy, 18(2): 1-5, 2023.

Abstract - The ability to decelerate is a key component of any successful rehabilitation program, yet it is often overlooked in favor of more traditional forms of rehabilitation and training. Deceleration, which is defined as the ability to reduce speed or momentum and stop or change direction, can be a key component of successful rehabilitation. The deceleration index is a new metric being used by some physical therapists and rehabilitation specialists to improve patient outcomes. The index is based on the principle that deceleration forces should match those created with acceleration. When patients can quickly and efficiently decelerate during physical activity, they are less likely to experience pain or injury. While the deceleration index is still in its early stages of development, there is promising evidence that it could be the missing link in effective rehabilitation. In this editorial commentary, we'll explore what the deceleration index is and why it is important to the rehabilitation process.

Sports Science Research Update

The effect of cam and pincer morphology on hamstring injury rates and return to play in elite young male football players. Acta Orthopaedica et Traumatologica Turcica, 57(1): 36-39, 2023.

Objective: This study aimed to determine the effects of cam and pincer morphology on the hamstring injury rate, treatment duration, and recurrence number of elite young male football players.

Methods: Seventy-five elite young male football players were included in this study. Participants were evaluated physically and radiologically for femoroacetabular impingement during pre-season or pre-transfer routine health examinations. Athletes' injury and treatment data for 6 seasons were then collected prospectively and evaluated retrospectively. Athletes were divided into 2 groups: patients with cam and pincer morphology (group 1) and those without (group 2). Both groups were compared regarding the number of hamstring injuries, treatment duration, and recurrence numbers.

Results: The mean age was 20.35 ± 1.37 in group 1 (n=23) and 18.92 ± 1.48 in group 2 (n=52). There was a significant difference between the 2 groups regarding age and body weight. The rate of hamstring injuries in group 1 was higher than in group 2, and this difference was statistically significant. Although the recurrence rate (group 1—35.7%, group 2—27.8%) and treatment duration (group 1: 18.4 ± 10.1 days, group 2: 12.1 ± 6.2) in group 1 were higher than in group 2, there was no statistical significance between 2 groups.

Conclusion: The present study has shown that hamstring injuries are more common in elite young male football players with cam and pincer morphology than those without. Therefore, further studies of musculoskeletal dysfunctions of femoroacetabular origin should be considered in the programs prepared to prevent and treat hamstring injuries.

Level of Evidence: Level III, Prognostic Study

Sports Science Research Update

The effect of probable COVID-19 infection on the National Football League player's performance and endurance during the 2020 season. Cureus, 15(3): e35821: 1-6, 2023.

Objective: The objective of this study was to assess whether the National Football League (NFL) players with probable coronavirus disease 2019 (COVID-19) during the 2020 season experienced a decline in athletic performance and endurance.

Methods: All players who were listed on the NFL's COVID-19 Injury Reserve (COVID-IR) list were screened for inclusion. Players were included in the study if they had spent ≥10 days on the COVID-19 IR list (which indicated a positive PCR test based on the NFL COVID-19 policies), had played in at least two games before and after going on the IR list, and primarily played an offensive or defensive position. The mean number of snaps played per game and Pro Football Focus (PFF) score per game were collected for each athlete, which served as surrogate measures of endurance and performance, respectively. The results were analyzed with players grouped by position, and then all players grouped as a whole. Within-group comparisons were performed via t-tests.

Results: A total of 78 players met the criteria for inclusion in the study. The overall mean PFF score pre-COVID-19 infection was 62.15 (SD: 6.93), while it was 61.73 (SD: 7.42) post-COVID-19 infection, showing a decrease of 0.42 after infection (n=78, p=0.33). The mean number of snaps played per game pre-COVID-19 infection was 38.99 (SD: 16.46) while it was 38.10 (SD: 17.05) post-COVID-19 infection, showing a decrease of 0.89 after infection (n=78, p=0.30). When grouped by position, statistically significant differences were seen with Defensive Backs' mean snaps played per game decreasing by 18.30 (n=6, p=0.03) and Defensive Linemen's mean PFF score decreasing by 3.77 points (n=21, p=0.03).

Conclusion: Based on our findings, COVID-19 infection negatively impacted endurance in Defensive Backs, and performance in Defensive Linemen. However, there was inconclusive evidence to show whether COVID-19 infection negatively impacted other positions when analyzed separately or all positions when analyzed together. Further studies with more participants are needed to fully assess the effects of COVID-19 on performance and endurance in elite athletes.

Sports Science Research Update

The good, the bad, and the ugly of consumer sleep technologies use among athletes: A call to action. Journal of Sport and Health Science, Ahead of Print: 1-3, 2023.

Introduction - For athletes, the quality and quantity of sleep are essential elements for optimizing recovery and subsequent performance. Unfortunately, athletes frequently face challenges getting sufficient, high-quality sleep. A recent systematic review and meta-analysis of 81 studies involving 1830 athletes concluded that compared to healthy non-athletes, athletes' sleep duration was shorter with poorer sleep efficiency. Notable sleep issues (e.g., sleep apnea and snoring, as well as insomnia and trouble falling asleep) were revealed in young athletes. It was common for sleep quality and architecture to change over time depending on the training period.

Highlights

- Consumer sleep technologies (CSTs) are the numerous software and hardware products that claim to monitor and even enhance sleep.
- At this time, CSTs cannot be used for the diagnosis or treatment of sleep problems.
- Orthosomnia is the name for a CST-driven obsession with getting the best possible sleep.

Sports Science Research Update

Validity of research based on publicly obtained data in sports medicine: A quantitative assessment of concussions in the National Football League. Sports Health, Ahead of Print: 1-10, 2023.

Context: Numerous researchers have leveraged publicly available internet sources to publish clinical research concerning incidence and recovery from injuries in National Football League (NFL) players.

Objective: This study aims to (1) provide a comprehensive systematic review of all publicly obtained data studies (PODS) regarding concussions in NFL athletes and (2) quantify the percentage of injuries identified by these studies in comparison with published concussion data from the NFL injury database.

Study Selection: A systematic review was conducted in accordance with PRISMA guidelines to identify all published studies utilizing publicly obtained data regarding concussions in NFL athletes.

Study Design: Systematic review.

Level of Evidence: Level 4.

Data Extraction: Manuscript details, factors related to the athletes of interest (eg, study period, positions included), and results (eg, concussion rate, number of total concussions, return-to-play data) were extracted independently by 2 authors. Results were compared with incident concussions reported from 2015 to 2019 by each medical staff member to the NFL database linked to the League's electronic health record (EHR).

Results: A total of 20 concussion-focused manuscripts based on PODS were identified from 2014 to 2020. PODS captured between 20% and 90% of concussions (mean, 70%) reported by medical staff to the injury database. PODS reported that 55% of concussions occurred on offensive plays, 45% on defensive plays and <1% occurred during special teams plays, compared with 44%, 37%, and 18%, respectively, as indicated by published data from the NFL injury database. When analyzed by position groups, running backs and quarterbacks comprised the most over-represented positions concussed in PODS, while offensive linemen, defensive backs, and linebackers comprised the most under-represented positions.

Conclusion: PODS captured approximately 70% of concussions reported by NFL medical staff to the NFL injury database. There is heterogeneity in the degree to which PODS were able to identify concussions, with a bias toward concussions among players at higher profile positions.